

## 07 Provenance

### Cryptographic Provenance Benchmark — api-oss-queue

**Company:** Anticloud FZ LLE

#### Receipt Generation Performance

Operation Time

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Single receipt creation 0.8ms

Chain verify (100 entries) 2.1ms

Chain verify (10K entries) 198ms

File hash (1MB) 4.2ms

File hash (1GB) 4,200ms

#### Chain Integrity Tests

Test Result

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Sequential append PASS

Tamper detection (modify entry 50 of 100) PASS — detected at entry 50

Reload from disk PASS — head matches

Cross-node receipt verification PASS — node fingerprint stable

#### Hash Algorithm

SHA3-256 (NIST FIPS 202). Collision resistance:  $2^{128}$ .

No MD5, no SHA1, no SHA2-256 (SHA3 only).